Fusang: Tardigrade Edition

Alignment-free phylogenetic inference with k-mer frequency vector cosine distances

INPUT

Unaligned FASTA (DNA or protein)

no alignment step

→

K-MER VECTORS

Spaced + contiguous k-mer frequency vectors

auto k,gap selection
multi-k ensemble: no k choice

→

COSINE DISTANCE

Pairwise cosine distance matrix

boundary classifier: homogeneous vs structured

→

TREE

NJ / FastME BIONJ+BNNI → Newick

simplified ≤500 taxa
DCM >500 taxa

Validated across simulation and real data

Indel-rich accuracy

n=200, indel rate 0.02, vs true tree (nRF ↓)

marginally better than FastTree2 (p=0.0507, borderline n.s.)

5% more accurate than IQ-TREE2 GTR ($p=7.9\times10^{-5}$)

Real-data validation

AFproject community benchmark, ETE3 nRF ↓

Fish mitogenomes (n=25, ~17 kb) **0.045**

community-benchmark ceiling — ties MSA+ML

13 mtDNA genes (168–1,866 bp) **0.451**

Fusang–FastTree2 agreement (13 loci)

SwissTree proteins (11 families) **0.329**

k-mer cosine vs Co-phylog 0.727 (p=0.0010)

no k selection at any scale: 500 bp genes → 17 kb genomes

Fast, scalable, ready to use

10,000 taxa **69.8 s**

single CPU core

1,000 taxa **5.2 s**

simplified pipeline

boundary classifier **150/150**

E2E scenario routing

pre-compiled Windows/Linux binaries · MIT license

github.com/zhanglknt/fusang-tardigrade